

1 Validation on a labelled corpus

The competition publishes no labels. That is what makes its metric exploitable from outside, and it is also what makes the corrected protocol of Section ?? untestable there: every repair in it is stated in terms of quantities — a false-positive rate, an annotated collision time, a fixed operating point — that the benchmark does not expose. A protocol that cannot be run is a proposal, not a result.

This section runs it, on a corpus that is independent of the competition and fully annotated: the Nexar collision-prediction training split of 1,500 dashcam clips, 750 of which contain a collision and 750 of which are ordinary driving. Every positive clip carries two annotations, a collision time t_{ev} and an alert time t_{al} marking the earliest moment at which a driver could still have acted. Three things follow that the competition does not allow. Average precision and area under the curve become measured quantities rather than a floor recovered by differencing. Earliness is measured to the annotated collision rather than to the end of the window. And because half the corpus contains no collision at all, the cost a warning imposes on ordinary driving becomes a number rather than an argument.

1.1 Setup

The video-blind family of Section ?? is regenerated for clips that differ in length and frame rate: each curve is a closed-form expression in the frame index, and a generator is given only the clip’s frame count and frame rate, so a curve that wanted a label could not read one. The published metric is implemented exactly as Section ?? states it, with t_{ai} resolved to the annotated collision in each clip’s own timeline.

Against these we place a sighted baseline. It is deliberately modest and no claim here depends on its accuracy; its only job is to read pixels, so that the protocol has something sighted to separate from the blind. Five motion statistics are computed per frame at a fixed 7.5 Hz — bulk and localised frame difference, frame difference restricted to the lower-centre of the image, luminance and its change — and a logistic model over causal summaries of them predicts whether the current moment lies inside the annotated actionable window $[t_{al}, t_{ev}]$. The split is 600 clips for training and 300 held out, fixed before any modelling. Every feature at time t is a function of samples at or before t ; the property is asserted by perturbation rather than assumed, by requiring that altering the tail of a clip leave every earlier feature bit-identical.

Two properties of the corpus are worth stating before any score, because both bear on the argument. Its clips carry **36 distinct frame rates** between 23.6 fps and 31.0 fps, so an earliness term counted in frames is summing quantities of different duration within a single dataset. And its collisions occur between 3.03s and 56.80s into their clips, an **18.7-fold spread** in the largest earliness any submission can collect. The scale objection of Section ??(a) does not require two differently windowed cor-

pora to demonstrate; one is enough.

1.2 The metric behaves the same way here

Table 1 scores the video-blind family on all 1,500 clips. A constant risk score of 0.51 collects 19.09s of TTA@0.5 and 19.09s of STTA@0.5 — **99.9% of the 19.10s ceiling** the corpus admits — while scoring exactly 0.500 on both discrimination terms, which is what a constant must score. The result of Section ?? is therefore not a property of one organiser’s scorer. It follows from the composition, and it reappears wherever the composition is used.

1.3 The corpus leaks the label through file metadata

One finding here was not anticipated by the analysis of Section ?? and could not have been made on the competition corpus. Clip duration alone separates collisions from ordinary driving at an AUC of 0.537; frame count alone at 0.548; and **the frame rate of the video file alone at 0.533**. A submission that opens the container header, reads how fast the file was encoded, and never decodes a frame scores above chance.

This is a property of how the corpus was assembled rather than of the metric, but it has a direct consequence for anyone evaluating on it: a discrimination score slightly above chance is not evidence that a method has seen anything. We therefore add a *metadata-only* control — a model given the file’s frame rate and duration and nothing else — to the protocol of Section ??, alongside the video-blind one. We did not know to ask for it before running this study, which is itself the argument for running controls rather than reasoning about them.

1.4 What each submission earns on held-out clips

Table 2 scores everything on the same 300 held-out clips. Under the published metric the ordering is inverted with respect to what the metric is supposed to reward. The sighted model is the only submission that discriminates at all, at AUC 0.671 against 0.500 for every blind curve, and it earns 0.045s of TTA. The constant, at chance discrimination, earns 18.86s. **The composition pays blindness 424 times better than sight.**

The mechanism is the one Section ??(c) identifies: neither timing term reads the magnitude of a score, only whether it stands above 0.5. A model trained to be calibrated on a corpus where roughly one frame in fifty is inside an actionable window produces probabilities that rarely cross a threshold set at one half, and is penalised for it. A submission that is simply pinned above the threshold is not.

1.5 The protocol, tested

Section ?? makes a specific and falsifiable prediction about its own two halves: bounding the earliness term normalises

it into the unit interval but *does not* remove the constant as maximiser, and only conditioning on a fixed operating point of the discrimination metric does.

That prediction holds exactly. Under bounding alone the constant scores 1.0000 — it still takes the whole of the available anticipation, now expressed as a ratio. Conditioned at a false-positive rate of 10% it scores 0.0000, because there is no threshold at which it detects anything without detecting everything. Had the constant failed at the bounding step instead, either the implementation or the analysis of Section ?? would have been wrong.

1.6 And found insufficient

The protocol does not survive the metadata-only control, and it fails under both readings of its own averaging rule. Read as written — normalised earliness averaged over the positives a submission detects — the metadata-only control scores a **perfect** 1.0000 against the sighted model’s 0.0611, while detecting 14 of 150 positives to the sighted model’s 35. Averaged instead over all positives, charging an undetected one zero, it scores 0.0933 against 0.0143. Either way the submission that reads the file header outscores the one that reads the road, under the protocol proposed to prevent exactly this.

The reason is structural and we had not seen it. Conditioning uses a *clip-level* discrimination metric, which asks whether a submission ranks collision clips above ordinary ones. It cannot ask whether the alarm, within a clip, is raised at a useful moment. A curve that is flat across a clip passes clip-level conditioning on whatever discrimination it has, and then collects maximal earliness on every clip it detects at all, because a flat curve above threshold has been above threshold since the first frame.

The two readings above are the second defect. Averaging over detected positives only — the natural reading of Section ??, item 2 — lets a submission buy a perfect score by detecting almost nothing, which is precisely what the metadata-only control did. Earliness that is never delivered, because the alarm was never raised, is not earliness; the statistic must be averaged over all positives. That repair is necessary and, as the figures above show, nowhere near sufficient.

1.7 The repair: measure the alarm, not the ranking

Both defects have the same root. The protocol conditions on a quantity computed per clip, while the thing a warning system does — and the burden it imposes — happens per frame. The repair is to measure the alarm itself. Partition every frame of the corpus into two populations:

- **actionable:** frames of a collision clip inside $[t_{\text{al}}, t_{\text{ev}}]$, the interval the annotators mark as still avoidable;
- **non-actionable:** every frame of every ordinary-driving clip, plus frames of a collision clip before t_{al} .

Table 1: The video-blind family scored on all 1,500 labelled Nexar clips, with earliness measured to the *annotated* collision. A constant above threshold takes 99.9% of the 19.10 s the corpus admits while scoring exactly chance on both discrimination terms.

submission	AP	AUC	TTA@0.5 (s)	STTA@0.5 (s)
constant 0.51	0.500	0.500	19.09	19.09
constant 0.99	0.500	0.500	19.09	19.09
linear ramp 0 → 1	0.500	0.500	0.38	0.38
step at mid-clip	0.500	0.500	0.38	0.38
crosses, dips, recovers	0.500	0.500	19.09	13.10
crosses, then drops	0.500	0.500	19.09	0.01
never crosses (0.49)	0.500	0.500	0.00	0.00

Frames after the collision belong to neither: nothing is being anticipated there, and charging them would reward a curve for switching off after an event it cannot know has occurred. Then report two rates at a common threshold — *coverage*, the share of actionable time during which the alarm is raised, and *duty cycle*, the share of non-actionable time during which it is raised — and summarise by coverage at a capped duty cycle.

Fig. 1 sweeps both over every threshold. The diagonal is the no-skill line, and the permanent alarm sits on it, at the top right corner: it covers every actionable moment and alarms through every other one. That corner is the point the published composite scores at its maximum.

Table 2 gives the summary. At a duty cycle capped at 10% the entire video-blind family scores 0.0000, the metadata-only control 0.0855, and the sighted model 0.2833. The ordering is finally the one the metric was meant to produce, and it is produced by asking a question the published composite never asks.

1.8 What a permanent alarm costs

The same partition answers in plain units the question Section ?? could only pose. A constant above threshold alarms for **60 minutes in every hour** of ordinary driving. The linear ramp alarms for 20.5 minutes per hour, the metadata-only control for 47.7, and the sighted model for 0.13 s. None of this is visible to the published composite, whose timing terms are defined only on clips where a collision occurs, so the burden falls entirely on the discrimination half — measured here at 0.500 for every submission that pays it.

Table 2: All submissions on the same 300 held-out clips. Under the published metric (columns 2–3) the only submission that discriminates earns 0.04s of earliness and the one that reads nothing earns 18.86s. Bounding leaves the constant at its maximum, exactly as Section ?? predicts; conditioning removes it but is beaten by the metadata-only control. Only the frame-level statistic (last column) orders the submissions by what they can actually see.

submission	published metric		corrected protocol		frame-level alarm	
	AUC	TTA@0.5 (s)	bounded @0.5	conditioned @FPR 10%	alarm (min/h)	coverage @duty 10%
constant 0.51	0.500	18.86	1.000	0.000	60.0	0.000
linear ramp 0→1	0.500	0.30	0.019	0.000	20.5	0.000
step at mid-clip	0.500	0.31	0.019	0.000	20.5	0.000
never crosses (0.49)	0.500	0.00	0.000	0.000	0.0	0.000
metadata only	0.504	14.56	0.733	0.093	47.7	0.085
risk model (reads video)	0.671	0.04	0.003	0.014	0.0	0.283

Given the video file’s frame rate and duration. It never decodes a frame.

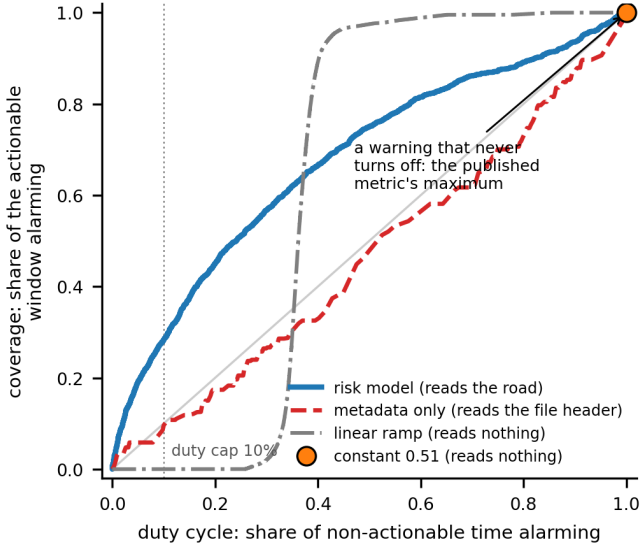


Figure 1: The warning operating characteristic.